

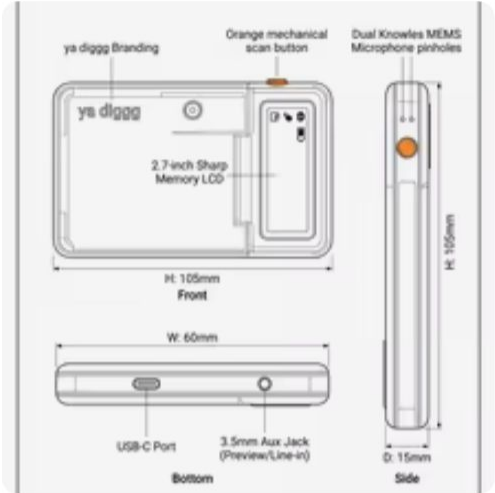


ya diggg: Official Tech Pack (Production -Ready)

Project Name: ya diggg
Revision: 2.0 (Sonic & WirelessIntegrated)
Origin: All-AmericanMade & Assembled

1. Three-View Drawing (Dimensions & External Ports)

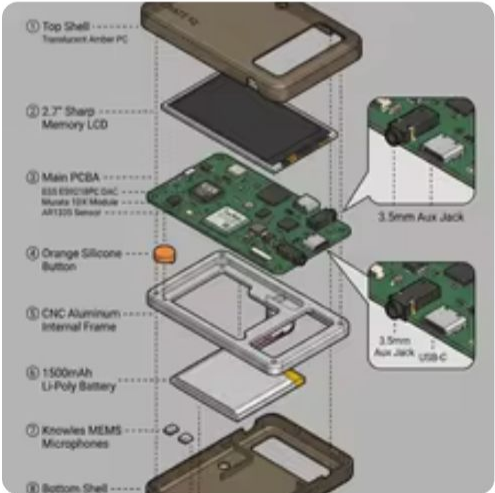
This drawingdefinesthe externalform factor, branding and user interfaceports.



Attribute	Specification
Dimensions	105mm (H) x 60mm (W) x 15mm (D)
Material	Smoke Amber TranslucentPolycarbonate(Grade: V0 Flame Retardant)
Finish	Matte Frosted Texture(Exterior) / Polished (InternalDetails)
Display	2.7-inch Sharp MemoryLCD (1-bit monochrome)
Branding	'ya diggg' laser-etched/ silkscreenedon top shell

2. Assembly Diagram (Exploded View & BOM)

This diagramdefinesthe internalstackingand componentrelationship



Bill of Materials (BOM)

#	Component	Material / Model	Qty
1	Top Shell	TranslucentAmberPC	1
2	Display Module	2.7" Sharp LS027B7DH01	1
3	Main PCBA	i.MX 8M Nano + ESS DAC + Murata 1DX	1
4	Scan Button	Orange Silicone (Soft-touch)	1
5	InternalFrame	CNC Aluminum (6061-T6)	1
6	Battery Pack	1500mAh Li-Polymer	1
7	Microphones	Knowles MEMS (High-SNR)	2
8	Bottom Shell	TranslucentAmberPC	1

3. Manufacturing Process Flow

Defined workflow for domestic "All-American" assembly



- CNC Machining:** Aluminum6061 internalframe (Heat sink & Chassis).
 - Injection Molding:** Smoke Amber Polycarbonateshells.
 - PCBA SMT:** High-precisionmountingof SoC, DAC, and MIPI Camera.
 - Integration:** Mountingcomponentsto internalframe.
 - Closure:** Sonic weldingor screw-fasteningof PC shells.
 - QC:** OCR accuracy, Sonic ID calibration, and Haptic LRA testing.
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4. Compliance & Safety Analysis

Applicable Regulations

- FCC Part 15 (U.S.):** Mandatoryfor the Murata 1DX wirelessmodule.
- RED 2014/53/EU (EU):** For internationalradio equipmentcompliance.
- RoHS/REACH:** Restrictionof hazardousmaterialsinelectronics.
- IEC 62368-1:** Safety standardfor audio/visualcommunicationtechnology.

Labeling Requirements

- "Assembled in USA":** Laser-etchedon the CNC Aluminumframe.
 - FCC ID:** Silkscreenedon internalPCB, visiblethroughtranslucentshell.
 - Recycling:** WEEE symboland Li-ion batterywarningsincludedon bottom shell.
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5. Engineering Notes

- Audio Path:** Ensure isolated ground plane for the ESS DAC to preventnoise from the Murata wirelessmodule.
- Haptics:** The LRA must be tuned for a 15ms "mechanicalclick" feel on scan trigger.
- Optics:** The 2.0mm pinhole aperturemust be perfectlyaligned with the AR1335 sensor.